

# AS 91173: Demonstrate Understanding of Electricity and Electromagnetism

## Topic Assessment: Level 2 Electrostatics

**Time Allowed:** 60 minutes. Write all answers on refill.

Use the following constants and formulas where appropriate:

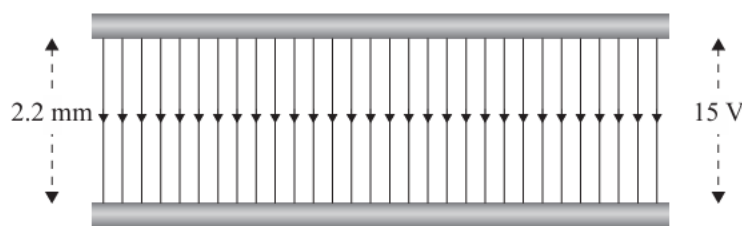
**Elementary Charge ( $e$ ):**  $1.60 \times 10^{-19} \text{ C}$     **Mass of an Electron ( $m_e$ ):**  $9.11 \times 10^{-31} \text{ kg}$

**Acceleration due to gravity ( $g$ ):**  $9.8 \text{ m s}^{-2}$

$$E = \frac{V}{d} \quad F = Eq \quad \Delta E_p = Eqd \quad E_k = \frac{1}{2}mv^2 \quad W = \Delta E_p$$

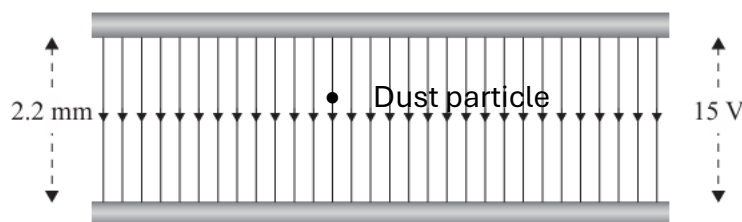
### Question One: Parallel Plates & Electric Fields

Two flat parallel conducting plates are connected across a 15V power supply. The distance between the plates is 2.2 mm.



- (a) State the direction of the electric field between the plates and show that the electric field strength is  $6.8 \times 10^3 \text{ V m}^{-1}$ .
- (b) Copy the diagram above and draw **at least four** electric field lines between the two parallel plates, including edge effects (fringing field).
- (c) An electron is released from rest at the surface of the **bottom (negative)** plate and accelerates towards the positive plate. Calculate the maximum speed with which the electron reaches the top plate. (Ignore gravitational effects).

(d) A small charged dust particle of mass  $m = 3.20 \times 10^{-12}$  kg is suspended at rest in the electric field between the plates.

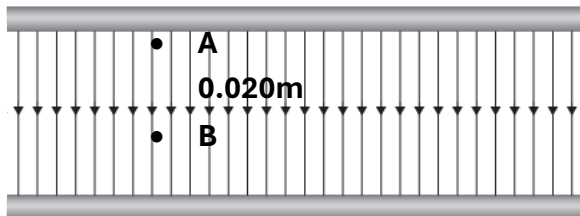


By applying physics principles:

1. Identify the charge on the dust particle (positive or negative).
2. Calculate the size of the charge on the dust particle.
3. Discuss the conditions required for the dust particle to remain suspended at rest.

## Question Two: Point Charges & Potential Difference

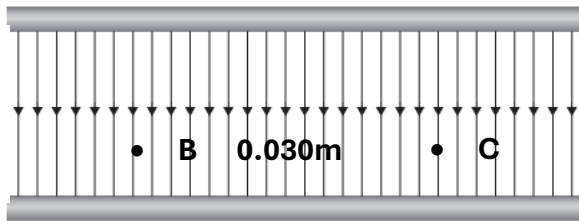
An ink-jet printing system uses parallel charged plates to deflect ink drops onto paper. Below is a diagram showing a region of uniform electric field with two points, **A** and **B**.



The uniform electric field strength between the plates is  $4.00 \times 10^4 \text{ V m}^{-1}$ . Point **A** is closer to the positive plate, and Point **B** is directly below Point **A**, separated by a vertical distance  $d = 0.020 \text{ m}$ .

- (a) Calculate the electrical potential difference (voltage drop)  $\Delta V$  between Point **A** and Point **B**.
- (b) A droplet carrying a charge of  $q = -4.80 \times 10^{-18} \text{ C}$  is moved from Point **A** to Point **B**. Calculate the change in electric potential energy ( $\Delta E_p$ ) of the droplet and state whether the droplet **gains** or **loses** electrical potential energy.

(c) The droplet is now moved horizontally from Point **B** to Point **C** through a distance of 0.030 m, as shown below:



Explain, using physics principles, why **no work** is done on the droplet when moving it from **B** to **C**.

(d) The distance between the original parallel plates is now **doubled**, while the total voltage source across the plates remains **unchanged**.

A student claims: *"Because the distance is doubled, an electron released from the negative plate will travel twice as far, so it will reach double the kinetic energy when it hits the positive plate."*

Explain whether the student's statement is correct or incorrect. In your answer, evaluate how changing the distance affects:

- The electric field strength ( $E$ ).
- The electrostatic force on the electron ( $F$ ).
- The work done ( $\Delta E_p$ ) on the electron across the entire journey.